Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Sonnenfeld, Gerald Ph.D.		
Project Title:	Bed Rest and Immunity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sonneng@binghamton.edu	**Fax:**	FY 607-777-2281
PI Organization Type:	UNIVERSITY	**Phone:**	607-777-4818
Organization Name:	Binghamton University, State University of New York		
PI Address 1:	Vestal Parkway		
PI Address 2:	P.O. Box 6000		
PI Web Page:			
City:	Binghamton	**State:**	NY
Zip Code:	13902-6000	**Congressional District:**	5
Comments:	PI moved to Binghamton University, from Morehouse School of Medicine, in 2004.		
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-05
Start Date:	07/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is 6/30/2007, per K. Major/NSBRI (12/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Butel, Janet (Baylor College of Medicine) Shearer, William (Baylor College of Medicine)		
Grant/Contract No.:	NCC 9-58-IIH00406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Exposure of humans to space flight conditions results in changes in immune responses. The contribution of these changes in the immune system to alterations in resistance to infection and tumors remains to be fully established. Chronic bed-rest with a head-down tilt has been a human model of choice for the effects of space flight on physiological systems. This model allows for no load-bearing on the legs of the subjects as well as a fluid shift to the head. Results utilizing the head-down tilt bed-rest model to study the effects of space flight conditions on the immune system are limited, but show a pattern similar to the results observed when humans are exposed to space. The head-down tilt bed-rest model has been established as valid for studying the effects of space flight conditions on functional dynamic immune responses such as cytokine production, but all of the studies to date have been carried out on men. The proposed study is designed to determine the effects of bed-rest on immune responses of women. The overall hypothesis to be tested is: maintenance of female subjects in the head-down tilt bed-rest model will result in suppression of functional immune responses and enhance susceptibility to infection. The specific aims of the proposed study are: to determine effects of maintenance of female subjects in the head-down tilt bed-rest model on functional immune responses. and to determine effects of maintenance of female subjects in the head-down tilt bed-rest model on resistance to infection. By carrying out these studies, we will correlate the effects of bed rest on immune function and control of viral infections. The French bed rest campaigns have been completed and the results are under anaylsis at the present time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many elderly individuals must remain in chronic bed rest and often develop decubiti (infected bed sores). The results of this study may aid in understanding mechanisms and in providing treatment for these patients. The analysis of the results of the study will provide new insight in this area as well as in the validity of the bed rest model for determining the effects of space flight conditions and countermeasures on immune responses of women.
Task Progress:	A first stage bed rest study has been carried out at the University of Texas Medical Branch, Galveston. Only three subnjects were involved and three additional subjects have been promised. The first and second campaigns of the main bed rest study with female subjects have been completed at the MEDES facility in Toulouse, France. The appropriate samples have been shipped to the United States and are currently under analysis. Additional analyses are underway regarding blastogenisis and cytokine production at the Universite Paul Sabatier in France. Preliminary data have been obtained on viral reactivation and on production of antibody against phiX-174.
Bibliography Type:	Description: (Last Updated: 04/28/2011)
Abstracts for Journals and Proceedings	Sonnenfeld G. "Cytokines and Interferons in the Space Flight Environment." ,2005 October. Cell Research. 2005 October;15:103. , Oct-2005
Abstracts for Journals and Proceedings	Sonnenfeld G. "Exploration class missions and return: Effects on the immune system." 21st annual meeting of the American Society for Gravitational and Space Biology, Reno, NV, November 2005. Gravitational and Space Biology 2005 Nov;19(1):24. , Nov-2005
Articles in Peer-reviewed Journals	Sonnenfeld, G., Aviles, H., Butel, J. S., Shearer, W. T., Niesel, D., Pandya, U., Allen, C., Ochs, H. D., Blancher, A., and Abbal, M. "Exploration class missions and return: effects on the immune system" Jan-2006
Awards	Sonnenfeld G. "Hunter Distinuished Lecture: Bed Rest and Immunity, Clemson University, January, 2006." Jan-2006